Independent Multi-Omics Target Discovery: Lung Adenocarcinoma (LUAD)

BIOT 6900 · Module 2 Assignment 2

1. Disease and data choice

I chose lung adenocarcinoma (LUAD) mainly because it is one of the few diseases where matched transcriptomic and proteomic data actually exist for the same patients, through the CPTAC (Clinical Proteomic Tumor Analysis Consortium) project. As someone without a strong coding background, I relied on AI-assisted coding support (Claude) to write and debug the Python pipeline, consistent with the computational scaffolding this course provides for biology students. My own contribution was in choosing the disease, understanding what each step of the pipeline was doing biologically, interpreting the results, and deciding what the findings do and do not support.

Three data layers were used:

Transcriptomics and proteomics came from the CPTAC LUAD cohort (213 patients: 111 tumor samples, 102 matched-normal samples), accessed through the cptac Python package. Both are open access. For each gene, a log2 fold-change and p-value were calculated comparing tumor to normal samples.

Genomic data came from the GWAS Catalog, searched for "lung adenocarcinoma" also open access. Each gene's strongest reported association across all studies was kept as its genomic evidence score.

Source:

Both sources are formally published: the CPTAC LUAD cohort comes from Gillette et al. (2020), and the GWAS Catalog is described in Sollis et al. (2023). I accessed both through their open-access interfaces rather than applying for any restricted data.

Because RNA and protein came from the same patients, those two layers are matched at the patient level. The GWAS data came from separate, unrelated cohorts, so it is not matched to the CPTAC patients. This means the final three-way comparison happens at the gene level rather than the patient level. the same category of integration used for the Alzheimer's exercise.

2. Method

The pipeline followed the course's standard steps: harmonize gene identifiers across the three files, check whether RNA and protein agree in direction, combine all three layers into one score per gene, and rank the results.

The most useful thing I learned in this process had nothing to do with writing code it was seeing, firsthand, why the "identifier problem" from the lecture is such a real issue. The CPTAC gene names came bundled with long database codes attached, and the GWAS file listed some genes in pairs joined by a dash when a genetic variant sat between two genes rather than inside one. Both had to be cleaned up before the three files could even be compared gene-by-gene, or real evidence would have been silently lost. This made the lecture's point about identifier harmonization click for me in a way that just reading about it wouldn't have.

After cleaning and merging the three files on gene symbol, 154 genes were present in all three datasets. For each of these genes, I looked at whether RNA and protein moved in the same direction (concordance), and then combined all three layers into one overall score, giving each layer equal weight.

1. Harmonization - "make sure all layers using the same name tag"
2. Concordance - "do RNA and protein agree on direction?"
3. Scoring - "combine three report cards into one grade"
4. Ranking - "sort everyone by their final score, biggest to smallest"

3. Results summary and biological interpretation

The table below summarizes the key numbers produced at each stage of the pipeline:

Metric	Value
CPTAC LUAD patients	213
Tumor samples	111
Matched-normal samples	102
RNA genes tested	59,286
Proteins tested	12,900
GWAS genes (gene-level)	361
Genes surviving 3-way harmonization	154
Concordant genes (RNA & protein agree)	112 (73%)
Discordant genes	42 (27%)
Top-ranked gene	FGFR2 (scored 0.944)
Known lung cancer genes recovered	FGFR2, CLPTM1L, CYP1A1

Of the 154 genes, 112 (73%) showed RNA and protein changing in the same direction. The top-ranked genes were:

Rank	Gene	RNA log2fc	Protein log2fc	GWAS -log10p	Concordant	Score
1	FGFR2	-1.72	-0.73	34.10	Yes	0.944
2	KRT8	1.47	0.64	14.70	Yes	0.883
3	USHBP1	-2.37	-1.28	8.70	Yes	0.881
4	GIPC2	-2.00	-1.13	7.40	Yes	0.847
5	CLPTM1L	1.06	0.47	28.00	Yes	0.842
6	BRCA2	0.96	-0.67	11.10	No	0.820
7	DNAJB4	-1.38	-0.80	7.40	Yes	0.817
8	CYP1A1	-2.52	-1.03	6.30	Yes	0.813

Full 15-gene list and all 154 scored genes are in targets_luad_all.csv.

I looked up the top hits to see whether any connected to known lung cancer biology, and two stood out.

CLPTM1L sits right next to TERT, the gene for telomerase, on chromosome 5. This region is one of the most frequently reported lung cancer risk regions in genetic studies. Because CLPTM1L and TERT are so close together

on the chromosome, it is not fully settled which of the two genes is actually responsible for the association. I think this is worth stating honestly rather than claiming more certainty than the field actually has. Still, recovering this locus from an independent dataset is a good sign that the pipeline picked up something real, not just noise.

CYP1A1 is an enzyme involved in breaking down chemicals from tobacco smoke. In doing so, it can actually turn some of those chemicals into more harmful, DNA-damaging forms this is one of the textbook explanations for how smoking leads to lung cancer at the molecular level. Seeing this gene near the top gave the result an actual biological story rather than just a number.

The top-scoring gene is FGFR2, which is a receptor for cell growth signaling and has been associated with a variety of cancers. The thing that was interesting for me in understanding FGFR2 is the fact that it has scored highest not because of the fact that it has shown up in any of the three layers in the strongest way but because of being strong in all of them simultaneously, which is the whole point of using this scoring approach. But there are two facts that should be discussed about this gene. First, the level of RNA and protein encoded by it is lower in the tumor cells than in the healthy ones, so this gene does not have an obvious role of being turned on during the development of cancer. Second, the GWAS score of FGFR2 is 34.10, which is the largest in my list, so this association can be caused by a general study of cancer instead of LUAD.

I also checked my top genes against **Open Targets** to see how they are regarded outside my own analysis. FGFR2, BRCA2, and CYP1A1 all have existing entries linking them to cancer, which was reassuring. What I did not find was any strong independent lung cancer association for USHBP1 or GIPC2, which ranked third and fourth in my list. I think that is worth flagging rather than glossing over, since it means those two are ranking highly on my combined score without much outside support behind them yet.

Non Obvious hits that's **worth a 2nd look** -**KRT8** was the second highest hit, and it is a structural protein found in epithelial cells, where lung adenocarcinoma develops. It is one of the few top proteins that are found in higher quantities in tumors compared to normal cases, which makes biological sense considering that tumors develop and restructure themselves. GIPC2 and DNAJB4 were two genes whose names were totally new to me. DNAJB4 is probably the one that makes sense, because it is a protein of a particular family of proteins responsible for correct protein folding, which is frequently seen in cancer research. As for GIPC2, I can say nothing about its function biologically.

4. A discordant gene: BRCA2

BRCA2 stood out because its RNA and protein moved in opposite directions RNA went up, protein went down. I think it is more likely a real biological pattern BRCA2 is a DNA-repair protein, and cells are known to tightly control DNA-repair proteins after they are made, since having too much or too little at the wrong time can be harmful. It's plausible that even though the cell increased RNA production for BRCA2, it also broke the protein down faster, so the RNA and protein levels ended up moving in different directions.

5. What this analysis can and cannot claim

FGFR2, CLPTM1L, and CYP1A1 are reasonable candidates worth investigating further; none of them are proven LUAD drivers based on this analysis alone. This analysis identifies genes with converging evidence, not confirmed disease-causing genes. A gene ranking highly here means it showed a consistent signal across three differen layers, independently collected types of data which is a stronger basis for further investigation than any one data type alone, but it is not the same as proving that gene causes lung cancer.

There is also a limit on what can be said about the different patient groups involved. Since RNA and protein came from the same 213 patients, I can say that within this group, protein levels for a gene are at least loosely connected to that same patient's RNA.

I cannot make the same claim about the genetic (GWAS) evidence, since it came from a completely different, unrelated group of people. So a high overall score reflects two true things about the same gene its behaviour in these tumours, and its association with disease risk in a different population is not one single observation about one group of people.

Finally, the GWAS layer was crucial becuase only 361 genes had usable genetic evidence, compared to all other genes in the RNA and protein data. This means a gene that didn't appear in my final ranked list might still be biologically important it may simply not have had genetic variant data available, rather than lacking real relevance to the disease.

Reference:

McKay, J. D., Hung, R. J., Gaborieau, V., Boffetta, P., Chabrier, A., Byrnes, G., … Brennan, P. (2008). Lung cancer susceptibility locus at 5p15.33. ***Nature Genetics, 40*(12), 1404–1406.** https://doi.org/10.1038/ng.254

Hukkanen, J., Pelkonen, O., Hakkola, J., & Raunio, H. (2002). Expression and regulation of xenobiotic-metabolizing cytochrome P450 (CYP) enzymes in human lung. ***Critical Reviews in Toxicology, 32*(5), 391–411.** https://doi.org/10.1080/20024091064273

Xie, L., Dang, Y., Guo, J., Sun, X., Xie, T., Zhang, L., Yan, Z., Amin, H., & Guo, X. (2019). High KRT8 expression independently predicts poor prognosis for lung adenocarcinoma patients. *Genes, 10*(1), 36. https://doi.org/10.3390/genes10010036